

## Experiment 14

### Problem Statement

Write a Python program using a class named *Complex* to add two complex numbers.

### Algorithm

Step 1: Start

Step 2: Create three objects:

*c1, c2, c3* of class *Complex*

Step 3: Input first complex number:

Read real and imaginary part

Store in *c1.real* and *c1.imag*

Step 4: Input second complex number:

Read real and imaginary part

Store in *c2.real* and *c2.imag*

Step 5: Compute sum:

$c3.real \leftarrow c1.real + c2.real$

$c3.imag \leftarrow c1.imag + c2.imag$

Step 6: Display result:

If  $c3.imag \geq 0$

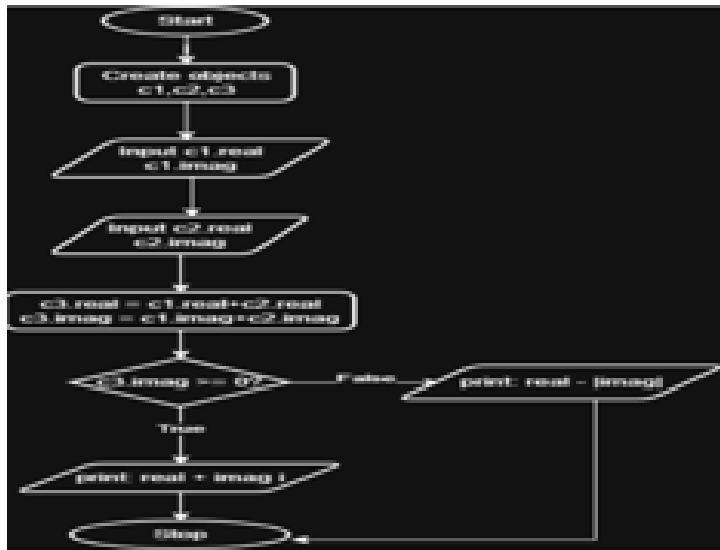
Print:  $c3.real + c3.imag \ i$

Else

Print:  $c3.real - |c3.imag| \ i$

Step 7: Stop

## Flowchart



## Execution

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### 14.1.1: Complex Number Addition using Class

Write a Python program using a class named Complex to add two complex numbers.

Create a class Complex with the following methods:

- `initComplex()`: Read user input for the real and imaginary parts.
- `sum()`: Compute the sum of two complex numbers and store the result in the current instance.
- `display()`: Display the complex number in the format "`x+yi`" or "`x-yi`".

The program should create three instances of Complex class: two for input complex numbers and one for the result. Initialize both complex numbers, compute their sum, and display the result.

**Input Format:**

- First line contains two space-separated numbers: real and imaginary parts of first complex number
- Second line contains two space-separated numbers: real and imaginary parts of second complex number

**Output Format:**

- Print a single line showing the sum of the two complex numbers in the format "`x+yi`" or "`x-yi`".

Sample Test Cases

```
1 class Complex
2     def __init__(self):
3         self.real = 0
4         self.imag = 0
5
6     def initComplex(self):
7         self.real, self.imag = map(int, input().split())
8
9     def sum(self, c1, c2):
10         self.real + c1.real = c1.real
11         self.imag + c1.imag = c1.imag
12
13     def display(self):
14         if self.imag >= 0:
15             print(f"{self.real} + {self.imag}i")
16         else:
17             print(f"{self.real} - {abs(self.imag)}i")
18
19
20 # Create three instances
21 c1 = Complex()
```

```
4
5 4
6 4
7 = 16
8
9 sum of two complex num is 16+0i
```